

Acoustic Emissions From Mechanical Keyboards: Analyzing Key Press Leakage of Loud Typists

Essentially: Loud Mechanical Keyboards Leak Whether a Key Is Pressed

Sudheendra Raghav Neela¹, Charizard Gyarados Primeape Trapinch²,
and Theresa Dachauer¹

¹ Institute of Quiet Studies, Sshhh-ville, Quietland

² ChatGPT, the internet

Abstract. When our (royal we) office dipshits Rannes and Holand³ press keys, their keyboards make sounds because they bought loud mechanical keyboards. If they had opted for silent switches, this research would be unnecessary. However, their choice of super loud switches, which seem to penetrate even the most advanced active noise-canceling headphones, has provided us with a unique opportunity to conduct a sophisticated inter-keystroke timing attack.

As the popularity of mechanical keyboards rises, particularly among employees like Rannes and Holand, understanding the auditory signatures associated with key presses become easier. We (royal we), without their permission, capture and analyze sound profiles generated during various typing scenarios, focusing on the correlation between specific key presses and their corresponding acoustic signatures. Our findings reveal that distinct sound patterns can be associated with individual keys, enabling us to identify not only the pressed keys but also the very users behind them — even in the cacophony of a bustling office environment.

By mounting our sophisticated inter-keystroke timing attack, we reconstructed the typing habits of these two "individuals" (if one can even call them that), revealing their innermost thoughts and musings, one keystroke at a time. This study underscores the implications of acoustic leakage for privacy and security, particularly in shared or public spaces, and suggests potential countermeasures to mitigate the risks associated with auditory eavesdropping. Ultimately, our results contribute to the broader discourse on information security in the digital age, emphasizing the urgent need for awareness of non-visual data leakage in our everyday technology use — because who knew that the sound of typing could be as revealing as the content itself?

Keywords:

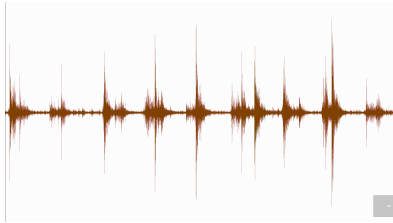
Mechanical Keyboards · Are · Loud ·
Rannes · Holand · Pls · Stop
Side-Channel Security

³ Names changed for privacy reasons

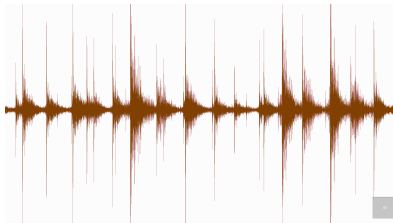
1 Related Work

In this section we discuss related work. There's been a lot of related work about keypress detection [7, 1, 5, 3]. There's also been a lot of related work about keystroke timing attacks [6, 4, 2, 8].

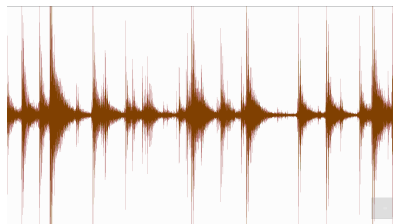
2 Samples



(a) Pictorial representation of an audio waveform of Cookie, the office dog, slurping water.



(b) Pictorial representation of an audio waveform of Rannes, an employee, typing on his keyboard.



(c) Pictorial representation of an audio waveform of Holand, an employee, typing on his keyboard.

Fig. 1: Pictorial representations of audio waveforms.

In this section, we present the waveforms of audio recordings capturing the typing sounds of Rannes and Holand. To establish a baseline for comparison, we

also include a waveform of Cookie, our office dog, as she enthusiastically slurps water from her bowl. It is worth noting that these recordings were obtained under questionable ethical standards, as the volume of their typing could arguably be considered a breach of auditory decorum.

In particular, we invite the reader to examine the pronounced fuzziness of the waveform spikes surrounding Holand’s keypresses, as illustrated in Figure 1c. In contrast, the waveform associated with Rannes’ keypresses, shown in Figure 1b, exhibits a notable lack of fuzziness. Lastly, we encourage you to extend a headpat to Cookie’s audio waveform presented in Figure 1a.

3 Reconstruction

In this section, we endeavor to reconstruct the text being typed by Rannes and Holand. To achieve this, we employ a sophisticated LLM hand-crafted algorithm. It is important to note that the reconstructed keypresses represent the most probable inputs rather than exact reproductions.

3.1 Analysis of Holand’s Typing

Holand appears to be typing the following:

P O R [Backspace] [Backspace] [Backspace]

He ceased typing abruptly upon realizing that we were recording his actions.

3.2 Analysis of Rannes’ Typing

Rannes appears to be typing the following:

In the shadows where the darkness dwells,

What the? This is some edgy stuff?

I rise from the ashes, breaking my chains,

Dude... stop. This sounds like some curse. HEY!

With a heart of fire and a soul of steel,

~~STOP IT RIGHT NOW~~

~~AM screen is damaged for the world! The echoes of
chaos a symptom of pain~~

Please! I'll let you use your mechanical keyboards!

~~REPAIRED~~

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